Case Study

Predictive Modeling for Alzheimer's Disease Onset Risk

Design Document

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1. Problem Understanding

Alzheimer's disease is a progressive brain disorder that affects memory, thinking ability, and daily functioning. Because symptoms often appear after significant brain damage has already occurred, early detection is crucial. Identifying individuals at risk early can help doctors begin treatments and lifestyle interventions that may slow cognitive decline and improve quality of life.

In this case study, the objective is to build a machine learning model that predicts whether a person is likely to develop Alzheimer's disease within the next five years. The prediction is based on cognitive test scores, demographic information, and brain MRI measurements.

The dataset used is from the OASIS-2 (Open Access Series of Imaging Studies) project, which contains longitudinal MRI and clinical data of older adults.

The model will learn patterns from this data to identify early signs of Alzheimer's risk. Key considerations include maintaining high sensitivity (recall) to detect potential cases early.

2. Dataset Understanding

Before performing exploratory analysis and building machine learning models, the dataset was inspected and cleaned to ensure it is suitable for modelling.

Initial Data Inspection
The dataset structure was first examined by checking column data types, dataset size, summary statistics, unique values, and the presence of missing values.

Standardizing Column Names
Column names containing spaces were standardized by replacing spaces with underscores. This ensures consistency and makes the dataset easier to use in analysis and automated pipelines.

Removing non-informative Features
The Hand column was removed because it contained only a single value for all records and did not provide useful information for prediction.

Creating the Target Variable
The Original Group column was converted into a binary target variable.

- Nondemented → 0
- Demented / Converted → 1

This transformation allows the problem to be treated as a binary classification task for predicting dementia risk.

Handling Missing Values
Missing values were found in SES and MMSE.

- SES values were filled using the median.
- MMSE values were filled using the mean.

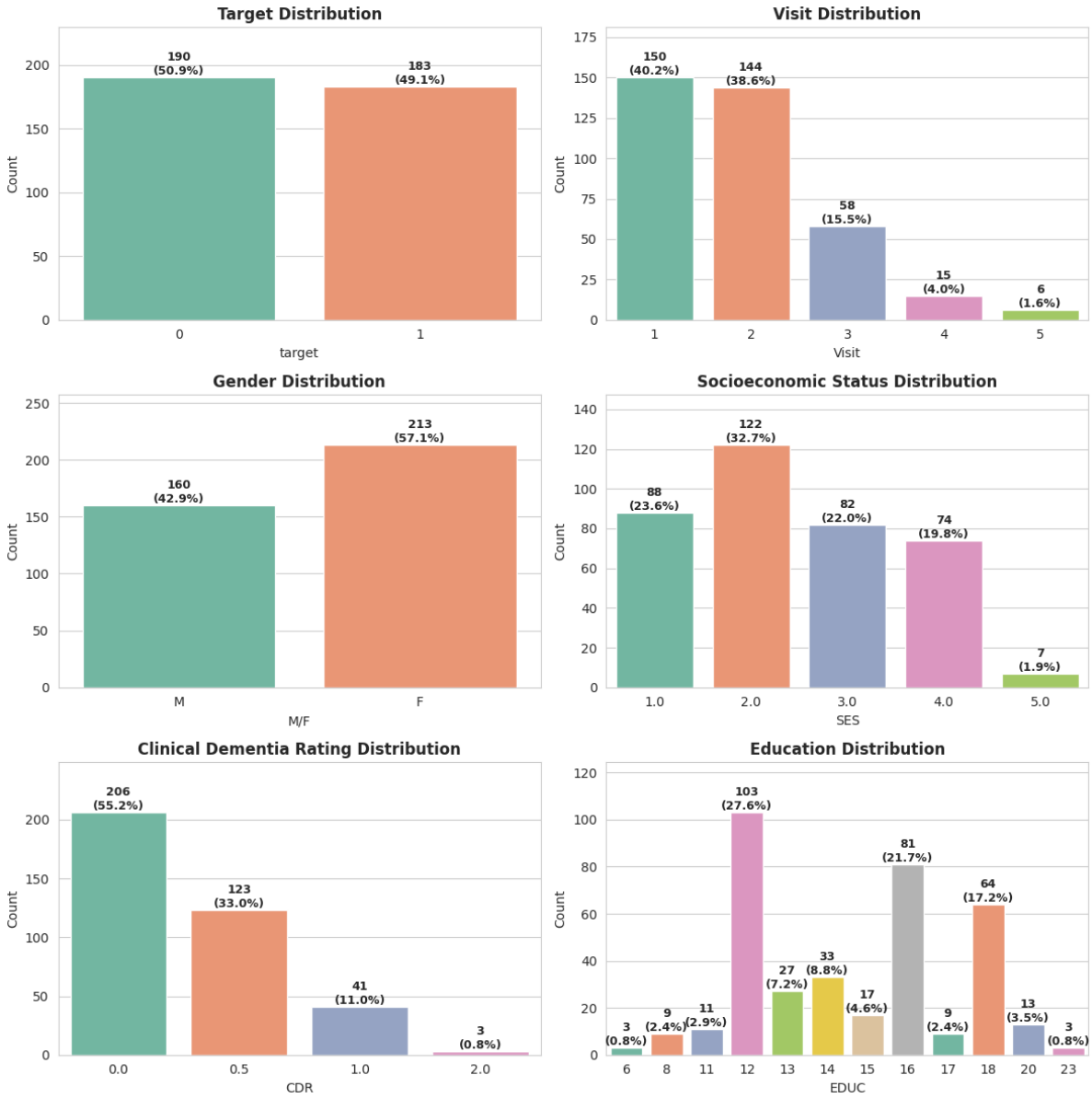
3. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is an important step in understanding the structure, distribution, and relationships within the dataset before building machine learning models.

3.1 Univariate Analysis

Univariate analysis focuses on understanding the distribution of individual variables in the dataset.

The following variables were analysed using count plots with percentage distributions:



Insights:

1. Target Distribution

- The dataset contains two major categories: Demented (1) and Nondemented (0) patients.
- The distribution of these two groups is almost balanced, which is beneficial for training classification models.
- A balanced dataset helps prevent model bias toward one class.

2. Visit Distribution

- The dataset contains multiple visits for several subjects, reflecting the longitudinal nature of the dataset.
- Visit 1 typically represents the baseline scan, while later visits capture follow-up observations.
- Most patients have only 1–2 visits (~79%).
- Very few have longitudinal follow-ups (visit ≥3)

3. Gender Distribution

- The dataset contains both male and female participants.
- Slight female dominance in dataset.

4. Socioeconomic Status Distribution

- SES values are distributed across multiple levels.
- Most participants belong to lower-to-middle SES groups (1–3).
- High SES (5) is extremely underrepresented.

5. Clinical Dementia Rating (CDR)

- Most individuals have CDR values of 0 or 0.5, indicating either no dementia or very mild dementia.
- Higher CDR scores represent increasing dementia severity.
- Majority ofa ptients are healthy or early-stage dementia.

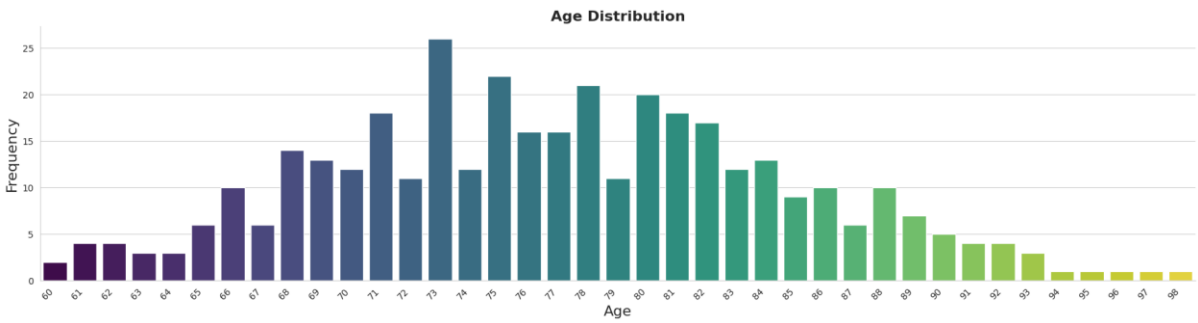
6. Education Distribution

- Peaks are typically observed around 12 years (high school) and 16 years (college education).
- Most participants have high school (12 years) or college education (16–18 years).

4.2 Bivariate Analysis

Bivariate analysis examines the relationship between two variables, particularly between predictors and the target variable.

Age Distribution Across Population



- Most participants fall between 70–85 years, indicating the dataset mainly represents an older population.
- The highest concentration is around 72–78 years, which appears to be the most common age group.

4.3 Outlier Detection

Outliers are extreme values that deviate significantly from many observations.

Two statistical techniques were used to detect outliers:

- Z-score method
- Interquartile Range (IQR) method

1. Z-Score Method

   This method identifies values that lie far from the mean of a distribution based on standard deviations.

   Insights:

   - A small number of outliers were detected in variables such as:
     - Visit, MR Delay, MMSE, CDR
   - Most other variables showed very few or no extreme outliers, indicating relatively stable distributions.

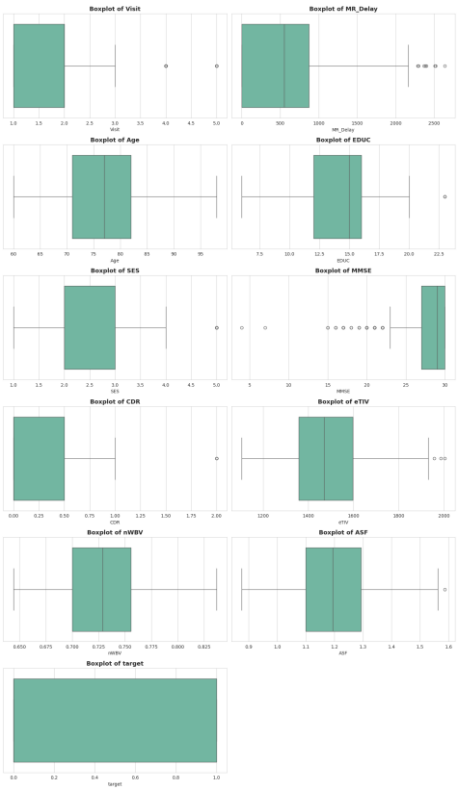
2. IQR Method

   The Interquartile Range (IQR) method detects outliers based on the spread between the 25th and 75th percentiles.

Insights

- The IQR method identified additional outliers in variables such as:
  - MMSE
  - SES
  - EDUC
  - eTIV
- MMSE showed the largest number of outliers, which may represent subjects with unusually low cognitive scores.

Boxplot Visualization



Insights:

- Boxplots visually confirmed the presence of outliers in certain features.
- Features such as MMSE and SES showed noticeable deviations from the main distribution.

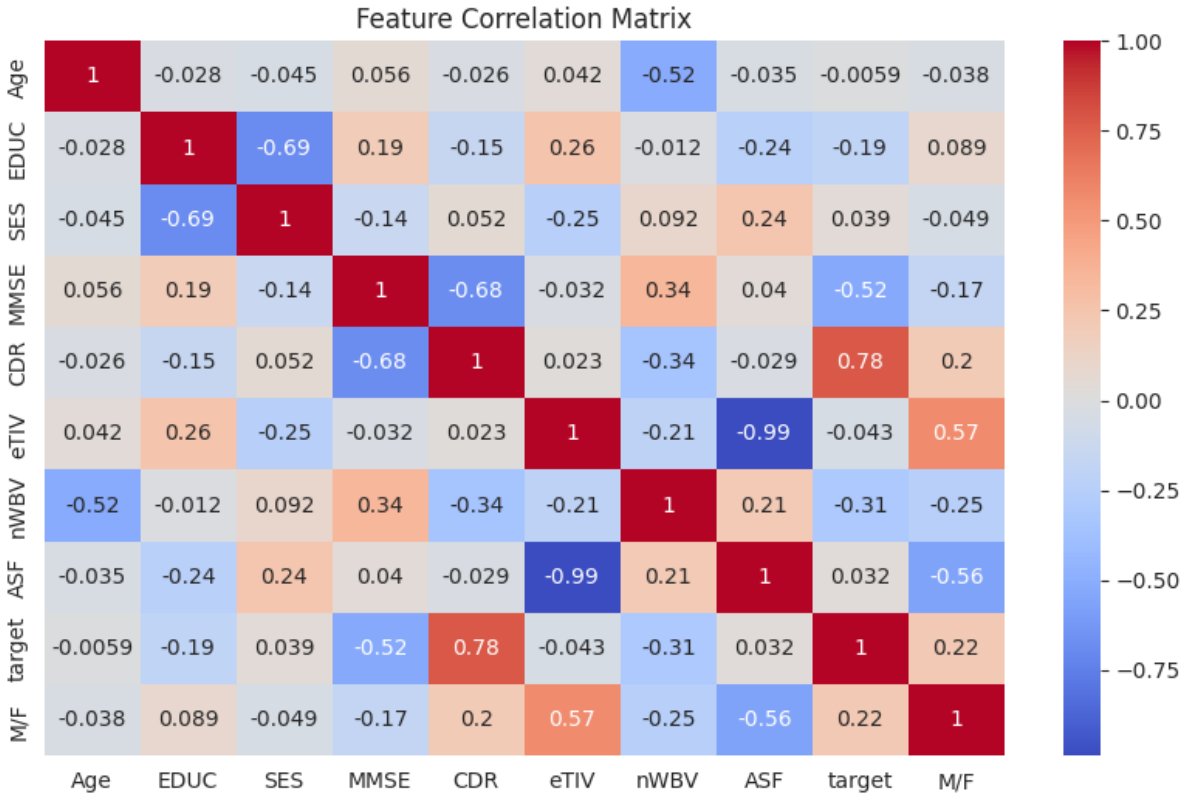
Distribution Plots

- Most variables show reasonable distributions, though some skewness is present in cognitive scores.
- Brain volume variables tend to follow more stable distributions.

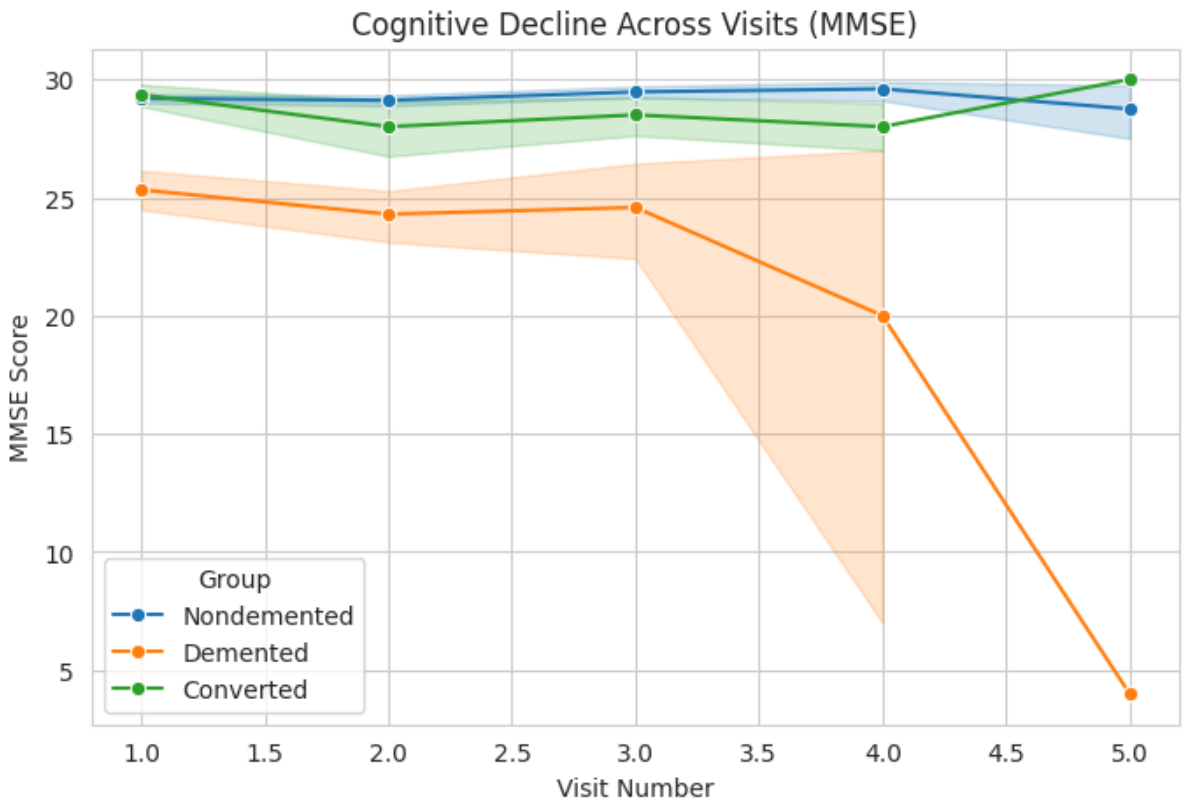
4.4 Correlation Analysis

Correlation analysis was performed to understand the linear relationships between numerical features.

A Pearson correlation matrix was generated to visualize how strongly variables relate to each other.

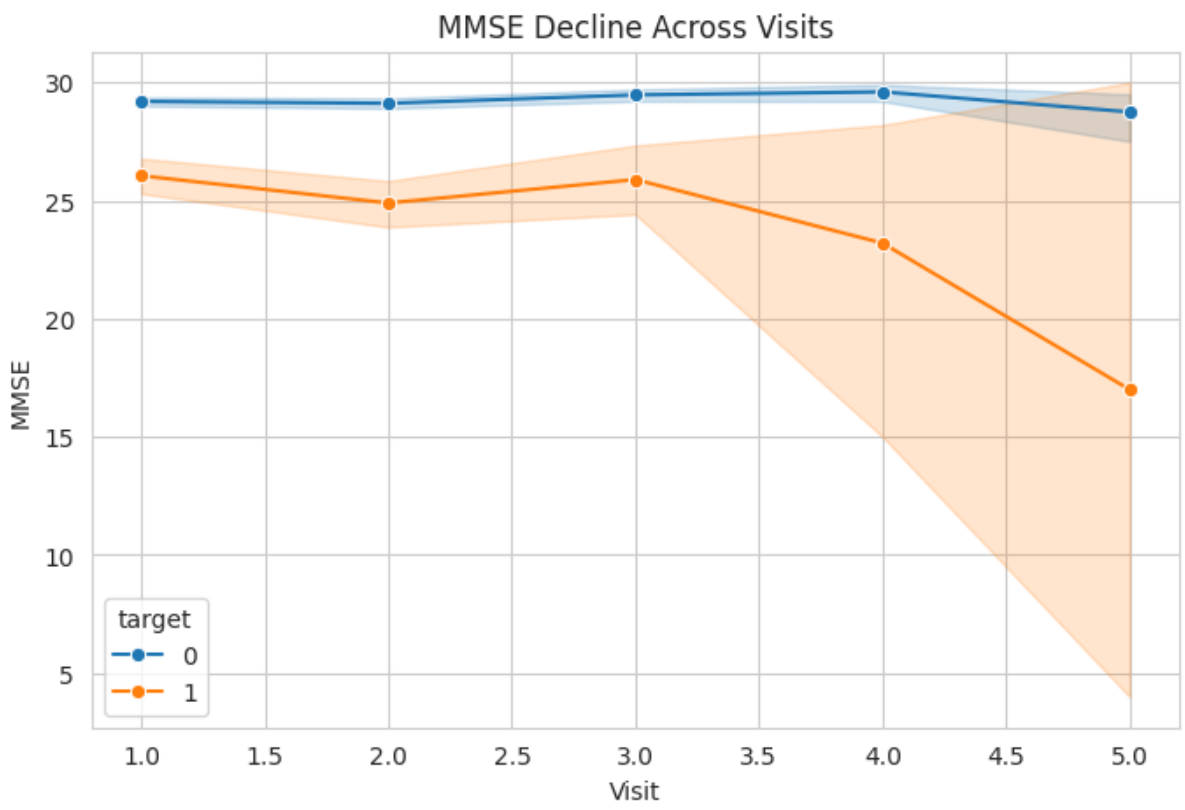


- CDR shows the strongest positive relationship with dementia, meaning higher CDR indicates higher dementia severity.
- MMSE is negatively related to dementia, so lower MMSE scores are linked to higher risk.
- Age is negatively correlated with brain volume, suggesting brain volume decreases with aging.
- MMSE and CDR are also strongly negatively related, meaning cognitive scores drop as dementia severity increase



- Nondemented individuals typically show stable MMSE scores across visits.
- Demented individuals show consistently lower MMSE scores.
- Some subjects demonstrate gradual decline, representing disease progression.

Brain Volume Decline Across Visits



- Brain volume tends to **decrease gradually over time**.
- Demented patients often show **faster decline in brain volume**

5. Model building & evaluation report

The models were evaluated using Accuracy, Precision, Recall, F1 Score, and ROC AUC to compare their ability to predict Alzheimer's risk.

Among the models, Logistic Regression achieved the highest recall (0.72), meaning it identified the largest number of potential dementia cases. This is important for early disease detection where missing a positive case can be critical. It also achieved the highest F1 score (0.57) among the models.

The Decision Tree model produced a slightly better ROC AUC (0.56) compared to the others, indicating somewhat better class separation, but its recall was lower than Logistic Regression. Random Forest showed recall like Logistic Regression (0.72) but had lower precision and overall performance. AdaBoost performed the weakest overall, with the lowest recall and F1 score.